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Design of Tri-Band Quadrature Hybrid coupler for WiMAX Applications

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Abstract—**As a quadrature hybrid is one important component in Butler matrix being a beamforming network for switched-beam antennas, this paper proposes a new design for quadrature hybrid coupler for WiMAX applications. To avoid the variation of different frequencies in different areas, the proposed design cover three frequency bands for WiMAX: 2.5 GHz (2.5-2.69 GHZ), 3.5 GHz (3.4-3.6 GHz) and 5.8 GHz (5.725-5.850 GHz). The proposed structure is relatively simple as it can be simply fabricated on single layer printed circuit board. The prototype is also constructed and tested to confirm its performance over the designated frequency bands.**

Keyword; Beamforming network, Butler matrix, quadrature hybrid coupler; three-branch-line (3-BL) quadrature hybrid; WiMAX.

I. INTRODUCTION

Currently, users in wireless communication systems require a high-speed data transmission to support modern applications such as video on demand, video conference and high speed down loading. WiMAX (Worldwide interoperability for Microwave Access) [1]-[2] is one wireless communication technology which can support high speed data transferring. This technology was developed from the WLANs or Wi-Fi based on IEEE802.16 standard which provides high speed Internet up to 75 Mbps. Its coverage is 30 miles or 50 kilometers. The operating frequencies have been assigned for three bands: 2.5GHz (2.5-2.69 GHz), 3.5GHz (3.4-3.6 GHz) and 5.8GHz (5.725-5.850 GHz). These frequency bands are allocated in different areas. To improve the performance of WiMAX systems, smart antennas are envisaged to be employed as they can give rise to Signal-to-Interference plus Noise Ratio while situating in an environment having co-channel interference and multipath. One typical type of smart antennas is switched-beam antennas. A Butler matrix being one type of beamforming network for switched-beam antennas have been considerably popular nowadays as it is simple but effective [3]-[5]. To support WiMAX operation without limitation of area, the beamforming network must be able to operate for all tri-ban: 2.5, 3.5 and 5.8 GHz.

From literatures, many attempts have been paid to developing some components in Butler matrix in order to fit in the designated applications. In the work presented in [6], a new branch-line hybrid coupler with arbitrary power division ratio has been proposed. Also the authors of the work presented in [7] have designed a compact multilevel folded-line RF coupler using two-level C-sections and branch-line hybrids with single and cascaded C-sections. This exhibits a significant reduction in foot print compared with the conventional design. In authors of the work presented in [8] have proposed introducing the open stubs at the symmetry-planes in order to avoid low-impedance line for two-arm branchline which can increase the operating bandwidth. The novel design right/left-hand miniaturized dual-band directional coupler has been shown in the work presented in [9]. Also, the work presented in [10] has shown the design including measured result for a dual band hybrid coupler employing rectangular patch for WLANs.

From literatures [6]-[10], we can see that the new design has been complex, hence manufacturing is impractical. Another one point is that they cannot support tri-band for WiMAX. Therefore, this paper presents tri-band quadrature hybrid coupler. This hybrid coupler can operate in three frequency bands for WiMAX: 2.5, 3.5 and 5.8 GHz. The proposed structure is not complex as it can be fabricated on single layer printed circuit board. Its prototype is constructed and tested to reflect the true performance covering three frequency bands for WiMAX.

The remainder of paper is organized as follows. Section II describes the design of proposed tri-Band quadrature hybrid coupler. It is for WiMAX applications covering frequencies at 2.5 GHz (2.50-2.69 GHz), 3.5 GHz (3.4-3.6 GHz) and 5.8 GHZ (5.725-5.850 GHz). Also, this section presents the fabricated prototype for the proposed hybrid. The prototype is tested and its measurement results are shown in Section III. Finally, Section IV concludes the paper.

II. DESIGN OF PROPOSED TRI-BAND QUADRATURE HYBRID

Quadrature hybrid couplers are 3 dB directional couplers with a 90° phase difference between the outputs of the through and coupled arms [11], and are often made in microstrip form. As shown in Fig. 1(a), all four ports are input (P1), direct (P2), coupled (P4) and isolated (P3). The behavior of quadrature hybrid coupler is as follows, when feeding the input signal at P1.

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- Signal will go to port P2 and P4 with equal power.
- Signals at P2 and P4 have phase difference of 90°.
- In ideal case, P3 is supposed to be isolated.

The optimal characteristic impedances can be found using graph of characteristic impedances Z_{a1}, Z_{a2}, Z_b in [12]. The characteristics impedances are $Z_0 = 50\,\Omega$, $Z_{a1} = 80\,\Omega$, $Z_{a2} = 121\,\Omega$, $Z_b = 37.4\,\Omega$ and the overall size of the main substrate is 44.17×20.02×1.67 mm^3.

Figure 1 shows the layout of the proposed hybrid. Its size and dimension can be found in Table I. Please note that the design is performed in CST Microwave Studio.

III. Measurement Results

After completing the design in last section, a prototype of coupler is constructed. The photograph of the proposed prototype is shown in Figure 2. This was fabricated on single layer printed-circuit board having dielectric constant of 4.8 and substrate thickness of 1.6 mm. Then, the prototype is tested using network analyzer to see its response in terms of S-parameters.

Figures 3 show the simulated compared with measured result return loss of proposed prototype. As we can see, the return loss less than -10 dB is obtained covering the designated frequency bands as follows. We obtain the return loss values of -10.143 to -10.327 dB, -20.090 to -15.249 dB, -13.451 to -16.871and the measured return loss values of -13.676 to -12.502 dB, -18.422 to -15.338 dB, -23.596 to-18.79 dB for 2.5, 3.5 and 5.8 bands, respectively.

Figure 4 shows the simulated and measured insertion loss of proposed prototype. Please note that these results are for the through port. We obtain approximately -1.145 to -1.285 dB, -0.119 to -0.197 dB, -2.712 to -3.237 dB from simulation and -4.959 to -1.168 dB, -1.45 to -3.188 dB, -4.003 to -4.114 dB from measurement for 2.5, 3.5 and 5.8 bands, respectively. For the coupled port shown in Figure 5, we obtain approximately -1.090 to -1.284 dB, -0.667 to -0.118 dB, -2.7119 to -3.436 dB from simulation and -4.393 to -2.24 dB, -3.982 to -2.135 dB , -6.75 4 to -4.895 dB from measurement for 2.5, 3.5 and 5.8 band, respectively.

The simulated and measured result for the isolation port of proposed prototype is shown in Figure 6. From the result, the simulated isolation values are -10.143 to -10.133 dB, -23.051 to -17.573 dB and 12.537 to -14.854 dB and also , from the measurement, -10.967 to -10.157 dB, -18.437 to -15.912 dB and -23.742 to -19.79 dB for 2.5, 3.5 and 5.8 bands, respectively.

Also the phase difference between through and coupled ports is very important. Figures 7-8 show that we obtain phase difference from simulation of 87.308°, -88.418°, -92.5369° and, also from measurement, -89.79°, -90.009°, -94.884° for 2.5, 3.5 and 5.8 bands, respectively.

In addition, Table II concludes the important values obtained from experiments. However, there is some deviation between the results in some cases because of some manufacturing error. As we can see, this confirms that the proposed hybrid can operate in three bands of WiMAX.

IV. Conclusion

This paper has proposed a new design for tri-band quadrature hybrid coupler for WiMAX applications. The designated band covers frequencies from 2.5 GHz (2.5-2.69 GHz), 3.5 GHz (3.4-3.6 GHz) and 5.8 GHz (5.725-5.850 GHz) for WIMAX applications. The reason for tri-band design is to cover all areas operating different frequency bands. The prototype of proposed hybrid is constructed and tested to confirm its performance throughout the designated band.

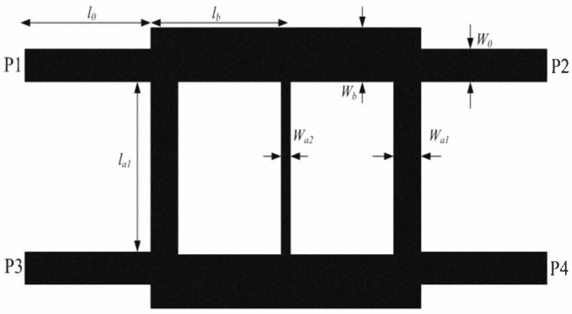

Figure 1. Layout of proposed tri-band hybrid.

TABLE I. Tri-band quadrature hybrid dimensions

Parameters	Size (mm)
W_b	5
W_{a1}	1.4
W_{a2}	0.1
W_0	3
l_{a1}	9.98
l_{a2}	9.98
l_b	9.98
l_0	13.14

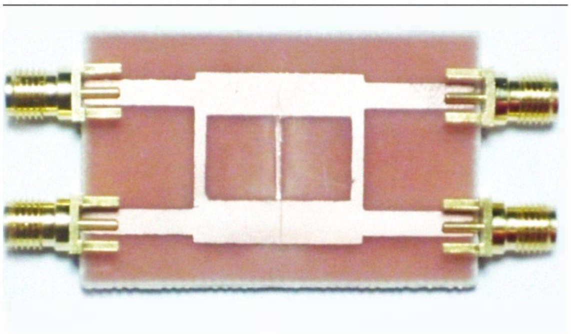

Figure 2. Photograph of the fabricated coupler.

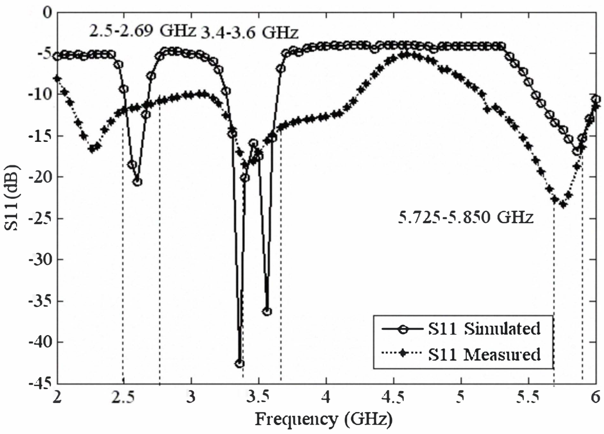

Figure 3. Measured and Simulated of return loss S_{11}(dB).

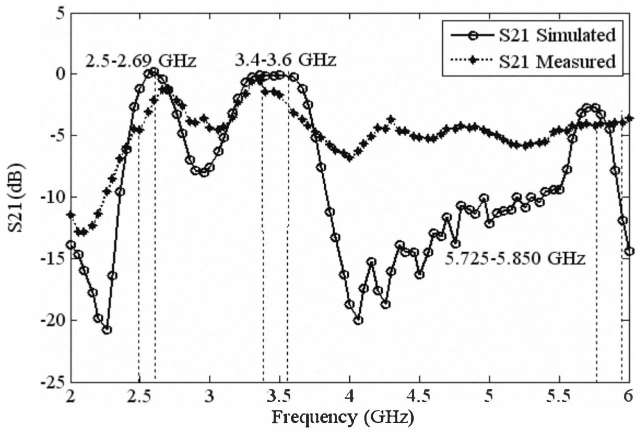

Figure 4. Measured and Simulated of insertion loss S_{21} (dB)

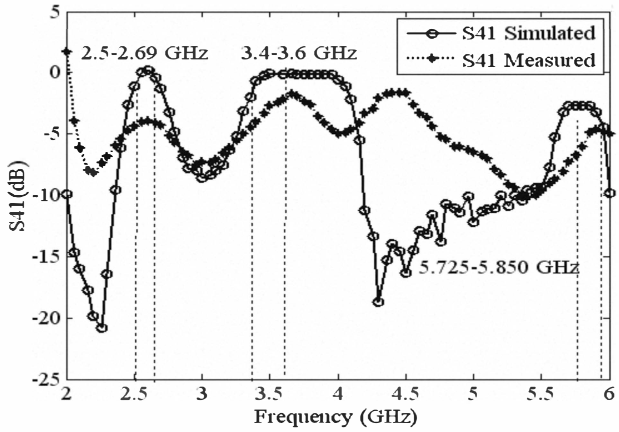

Figure 5. Measured and Simulated of coupled port S_{41} (dB)

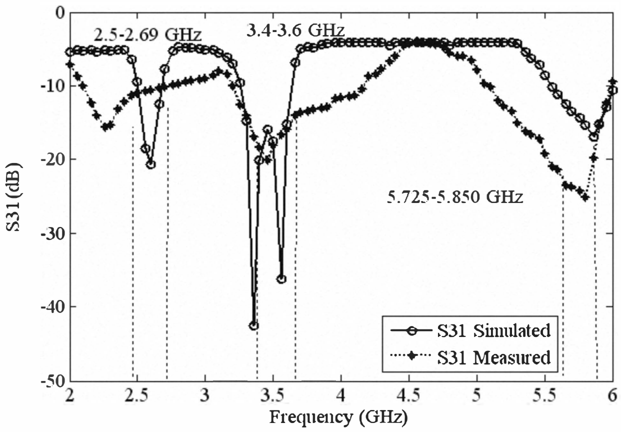

Figure 6. Measured and Simulated of isolation S_{31}(dB)

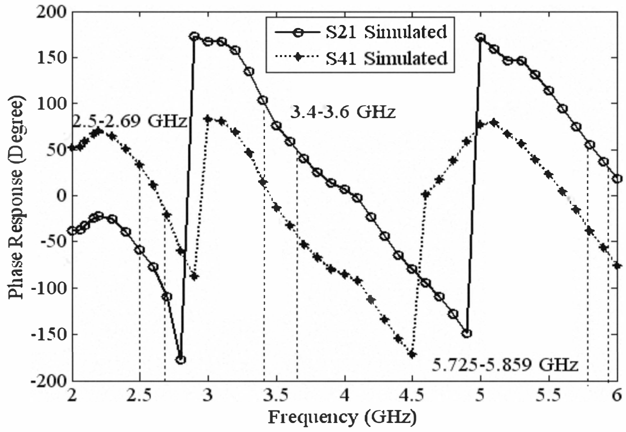

Figure 7. Simulated phase response between S_{21} and S_{41} (dB)

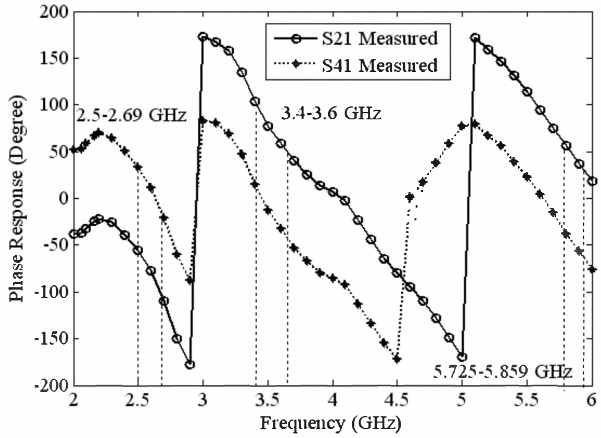

Figure 8. Measured phase response between S_{21} and S_{41} (dB)

TABLE II. MEASURED PERFORMANCE OF TRI-BAND QUADRATURE HYBRID COUPLER

Frequency	2.5 GHz	3.5 GHz	5.8 GHz
Insertion loss (S_{21}) Simulated	-1.145 dB	-0.0856 dB	-3.237 dB
Coupled port (S_{41}) Simulated	-1.145 dB	-0.0412 dB	-2.711 dB
Insertion loss (S_{21}) Measured	-4.595 dB	-1.684 dB	-4.072 dB
Coupled port (S_{41}) Measured	-4.393 dB	-1.684 dB	-6 dB
Phase (S_{21}) Simulated	36.727°	-8.413°	-33.9139°
Phase (S_{41}) Simulated	-50.581°	80.005°	58.623°
Phase (S_{21}) Measured	33.954°	-12.587°	-37.752°
Phase (S_{41}) Measured	-55.836°	77.422°	56.132°

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